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lab 1-4 week 6 ADVANCED DATABASE

LAB 1:

code:

```
Query Query History
1 WITH RankedFilms AS (
2     SELECT
3         c.name AS category_name,
4         f.title,
5         f.length,
6         DENSE_RANK() OVER (PARTITION BY c.name ORDER BY f.length DESC) as length_rank
7     FROM
8         film f
9     JOIN
10        film_category fc ON f.film_id = fc.film_id
11    JOIN
12        category c ON fc.category_id = c.category_id
13 )
14 SELECT *
15 FROM RankedFilms
16 WHERE length_rank <= 3;
```

result:


```
Query Query History
1 WITH MonthlyRevenue AS (
2     -- Calculate total revenue per month
3     SELECT
4         DATE_TRUNC('month', payment_date) AS payment_month,
5         SUM(amount) AS current_revenue
6     FROM
7         payment
8     GROUP BY
9         DATE_TRUNC('month', payment_date)
10 ),
11 RevenueWithLag AS (
12     -- Use LAG() to get the previous month's revenue
13     SELECT
14         payment_month,
15         current_revenue,
16         LAG(current_revenue, 1) OVER (ORDER BY payment_month) AS previous_revenue
17     FROM
18         MonthlyRevenue
19 )
20 -- Calculate the Month-over-Month growth percentage
21 SELECT
22     payment_month,
23     current_revenue,
24     previous_revenue,
25     ((current_revenue - previous_revenue) / previous_revenue) * 100 AS growth_percentage
26 FROM
27     RevenueWithLag;
```

result:

Query Query History

1 WITH MonthlyRevenue AS (
2 -- Calculate total revenue per month
3 SELECT
4 DATE_TRUNC('month', payment_date) AS payment_month,
5 SUM(amount) AS current_revenue
6 FROM
7 payment
8 GROUP BY
9 DATE_TRUNC('month', payment_date)
10),
11 RevenueWithLag AS (
12 -- Use LAG() to get the previous month's revenue

Data Output Messages Notifications

Showing rows: 1 to 4

	payment_month timestamp without time zone	current_revenue numeric	previous_revenue numeric	growth_percentage numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.0035632866530000
3	2007-04-01 00:00:00	28559.46	23886.56	19.56288389789069669300
4	2007-05-01 00:00:00	514.18	28559.46	-98.19961581906660700200

LAB 3:

query query history

```
1 SELECT
2     customer_id,
3     payment_date,
4     amount,
5     SUM(amount) OVER (
6         PARTITION BY customer_id
7         ORDER BY payment_date
8     ) AS customer_lifetime_value
9 FROM
10    payment;
```

Data Output Messages Notifications

Showing rows: 1 to 1000

	customer_id smallint	payment_date timestamp without time zone	amount numeric (5,2)	customer_lifetime_value numeric
1	1	2007-02-14 23:22:38.996577	5.99	5.99
2	1	2007-02-15 16:31:19.996577	0.99	6.98
3	1	2007-02-15 19:37:12.996577	9.99	16.97
4	1	2007-02-16 13:47:23.996577	4.99	21.96
5	1	2007-02-18 07:10:14.996577	4.99	26.95
6	1	2007-02-18 12:02:25.996577	0.99	27.94
7	1	2007-02-21 04:53:11.996577	3.99	31.93
8	1	2007-03-01 07:19:30.996577	4.99	36.92
9	1	2007-03-02 14:05:18.996577	3.99	40.91
10	1	2007-03-03 16:00:04.996577	0.99	41.90

LAB 4:

code:

a,

```
WITH CustomerSpend AS (
    -- Step 1: Aggregate total spend by Customer and Country
    SELECT
        co.country,
        c.customer_id,
        c.first_name,
        c.last_name,
        SUM(p.amount) AS total_spend
    FROM
        customer c
    JOIN payment p ON c.customer_id = p.customer_id
    JOIN address a ON c.address_id = a.address_id
    JOIN city ci ON a.city_id = ci.city_id
    JOIN country co ON ci.country_id = co.country_id
    GROUP BY
        co.country, |
        c.customer_id,
        c.first_name,
        c.last_name
),
```

b,

```
RankedCustomers AS (  
  -- Step 2: Apply RANK() partitioned by Country  
  SELECT  
    country,  
    first_name,  
    last_name,  
    total_spend,  
    RANK() OVER (PARTITION BY country ORDER BY total_spend DESC) as spend_rank  
  FROM  
    CustomerSpend  
)
```

c,

```
-- Step 3: Select where rank = 1  
SELECT  
  country,  
  first_name,  
  last_name,  
  total_spend  
FROM  
  RankedCustomers  
WHERE  
  spend_rank = 1;
```

result:

Data OutputMessagesNotifications

SQL

Showing rows: 1 to 108

	country character varying (50)	first_name character varying (45)	last_name character varying (45)	total_spend numeric
1	Afghanistan	Vera	Mccoy	67.82
2	Algeria	June	Carroll	151.68
3	American Samoa	Anthony	Schwab	47.85
4	Angola	Claude	Herzog	93.80
5	Anguilla	Bobby	Boudreau	99.68
6	Argentina	Jordan	Archuleta	129.71
7	Armenia	Stephanie	Mitchell	118.75
8	Austria	Nora	Herrera	113.74
9	Azerbaijan	Raymond	Mcwhorter	108.75
10	Bahrain	Seth	Hannon	108.76
11	Bangladesh	Michelle	Clark	146.68
12	Belarus	Clara	Shaw	189.60
13	Bolivia	Joel	Francisco	92.78
14	Brazil	Marion	Snyder	194.61
15	Brunei	Lois	Butler	107.66
16	Bulgaria	Tyrone	Asher	112.76
17	Cambodia	Elmer	Noe	92.76
18	Cameroon	Albert	Crouse	96.78
19	Canada	Curtis	Irby	167.62
20	Chad	Javier	Elrod	122.72
21	Chile	Janet	Phillips	124.74